

Anthony Hughes

✉ ajhughes3@sheffield.ac.uk

🌐 anthonyhughes.github.io

🐙 @anthonyhughes

☎ +447496860312

I am a **PhD student** focusing on **privacy in language models** with publications at **EMNLP** and **EACL**.

Prior to my PhD, I spent **10 years in industry** as a software engineer. Currently, *I'm taking my research experiences in **privacy and pivoting towards technical safety research***. As a **SPAR fellow**, I'm working on detecting whether a model has been **backdoored** through **poisoned** training data.

My PhD research develops algorithms for **protecting the privacy of individuals** in language models. This has involved **mid- and post-training techniques** for preserving privacy in settings such as summarisation, followed by **circuit-based interpretability** approaches to understanding and mitigating personal information leakage. This work uses **mechanistic interpretability as a tool** to understand how models **memorise** and surface sensitive information with **formal guarantees** from differential privacy.

The focus right now is applying these technical skills to **AI safety**. In **SPAR**, I'm studying the **advantages attackers gain from poisoned training data**, while my independent work examines the **vulnerability of safety-critical components like classifiers and probes**. For example, my recent project uses **membership inference methods** from my privacy research to show that ***fine-tuned safety classifiers can leak information about their training data***, directly connecting privacy attacks to safety concerns.

Education

- **PhD Computer Science**
 - University of Sheffield, 2024-*Current (estimated completion April 2027)*
 - *Courses: Time-series Analysis, Machine Learning, Speech Processing and Technologies, Text Processing, and NLP.*
- **PgDip Speech and Language Technologies - Leadership**
 - University of Sheffield, 2023-2024
- **MSc Computational Linguistics (Distinction)**
 - University of Wolverhampton, 2021-2023
- **BSc Computer Science (1st Class, Hons)**
 - Nottingham Trent University, 2009-2013

Research Experience

- **Research Internship | June 2026 - July 2026 |**
 - Collaborators/Mentors: **Jacob Imola, Gautam Kamath (University of Waterloo)**
 - Project: ***Designing and Developing New Neural Architectures for Unlearning***
- **Research Internship | March 2026 - May 2026 |**
 - Mentors: **Andrew Draganov (Arcadia/LASR Labs)**
 - Project: ***Detecting Whether an LLM has Been Poisoned***
- **Visiting Graduate Researcher | July 2025 - September 2025 |**
 - Collaborators/Mentors: **Vasisht Duddu, N. Asokan (University of Waterloo)**
 - Accepted EACL: ***PATCH: Mitigating PII Leakage in Language Models with Privacy-Aware Targeted Circuit Patching***

Selected Publications

- **PATCH: Mitigating PII Leakage in Language Models with Privacy-Aware Targeted Circuit Patching**
 - EACL 2026 | Findings
 - <https://arxiv.org/abs/2510.07452>
- **How Private are Language Models in Abstractive Summarisation?**
 - EMNLP 2025 | Main Proceedings
 - <https://aclanthology.org/2025.emnlp-main.1531/>

Example Safety Project

- **Safety at What Cost? Privacy Attacks on Fine-Tuned Safety Classifiers**
 - <https://github.com/anthonyhughes/dp-mental-health>

Industry

- **Data Language | Jan 2014 - September 2023**

Junior Engineer → Lead Software Engineer → NLP Engineer

- Working as a Lead Software Engineer delivering data and software solutions for a wide range of clients.
- Built an automated text content classification SaaS product, enabling clients easy access text analytics. Led the development of a data visualisation tool allowing customers to view classification quality metrics.
- Integrated language models into a SaaS data platform, enabling clients to query their graph data using natural language. Surfaced client graph data for LLM interactions, facilitating more intuitive client engagement with their stored data.

- **Ontoba | June 2013 - June 2014**

Software Engineer working on services solving data silo issues with graphs/linked data focussed solutions.

- **Press Association | July 2011- September 2012**

Internship (Software Engineer) working on a new digital platform centered around semantic web technologies.

Mentoring Experience

- **Fatemeh Honarvar Nazari | Jan 2026 → Present**

Project: *Understanding how models reason about private information.*

- **Sajad Rahmanian Ashkezari, Neel Sanjaybhai Faganiya, Lucas Kopp | Sep 2025 → March 2026**

Project: *How Do Language Models Encode Privacy Norms?*

- **Yangming Cao | Feb 2025 -> Sept 2025**

Project: *Clinical Coding of Medical Texts*

- **Emma Ellwood | Apr 2025 -> Sept 2025**

Project: *Obfuscation of Gender and Familial Information in Medical Summaries*

Talks

- **EurIPS | December 2025 | Foundations of Language Model Security**

Title: *PATCH: Mitigating PII Leakage in Language Models with Privacy-Aware Targeted Circuit Patching*

- **EMNLP | November 2025 | Main Conference**

Title: *How Private are Language Models in Abstractive Summarization?*

- **Insigneo Institute | May 2024 | Synthetic Data Workshop**

Title: *Identifying and Aligning Medical Claims Made on Social Media with Medical Evidence*

Grants/Funding

- **Vector Institute | Jun 2026 → July 2026**

Title: *Privacy Guarantees for Third-Party Model Auditing.*

- **NIHR/BRC Sheffield | APR 2025 → SEPT 2025**

Title: *AI-driven Medical Record Redaction during Adoption and Gender Reassignment in Primary Care*

Academic Service

- **Reviewer, Call For Talks, EurIPS Foundations of Language Model Security | 2025**
- **Reviewer, Student Research Workshop, EACL | 2026**